

Supplementary Material D: Implementation Fidelity

Observation Checklist, Scoring Rubric, Session Coverage,
Observed Scores, and Deviation Log

Note. This document was compiled from the independent observer’s session records, the facilitator session logs, and weekly supervision minutes (August 2026). It is anonymized for peer review. The checklist was developed by the principal investigators prior to data collection and was used both for pre-intervention facilitator competency assessment (mean score 84%, range 81–88%; see Supplementary Material E) and for in-study observation. The control condition was verified separately using the absence check described in Section D5 and in Supplementary Material B (Section B5).

D1. Fidelity Monitoring Procedures

Implementation fidelity was monitored through three complementary mechanisms:

1. **Facilitator session logs.** Each of the four experimental facilitators completed a structured log after every session (20 logs per facilitator; 80 total), recording: (a) activities conducted in each of the five protocol phases, (b) time actually allocated to each phase, (c) AI tools used and whether the GitHub commit was completed, (d) any deviations from the planned sequence, and (e) a brief qualitative note on group engagement. Logs were reviewed at weekly supervision meetings.
2. **Weekly supervision meetings.** The four facilitators met with the principal investigators for approximately 45 minutes each week throughout the intervention period. Meetings served to review session logs, address implementation questions, and recalibrate facilitator practices when logs indicated drift from the protocol. Meeting minutes were recorded by the principal investigators.
3. **Independent observer structured observations.** An independent observer—a researcher not otherwise involved in the study—attended a stratified random selection of 4 of the 20 experimental sessions (20%), distributed to ensure that each of the four facilitators was observed at least once and that both early (Weeks 1–5) and late (Weeks 6–10) sessions were covered. The observer used the 15-indicator checklist described in Section D2 and the scoring rubric described in Section D3. Additionally, 2 control sessions were observed to verify treatment differentiation (Section D5).

The *a priori* threshold for adequate fidelity was set at 80% of the maximum possible score per session, consistent with thresholds employed in comparable educational intervention research (Clemens et al., 2025; Festiyed et al., 2026).

D2. The 15-Indicator Observation Checklist

The checklist comprises 15 indicators organized across the three components of the connectivist framework: Networked Instructional Design (NID; indicators 1–5), Technological Mediation (TM; indicators 6–10), and Connectivist Assessment (CA; indicators 11–15). Each indicator is rated on a three-point scale (Section D3). Table D1 presents the full checklist as used by the independent observer.

Table D1: Full 15-indicator structured observation checklist. Scoring: Present = 1.0; Partially present = 0.5; Absent = 0. Session fidelity score = sum / 15.			
No.	Indicator	Observable evidence expected	Component
1	Teams are composed of students from at least two distinct academic disciplines and contain between four and six members.	Observer can verify team composition from the roll or direct inquiry; no team is mono-disciplinary or exceeds six or falls below four members.	NID
2	The session opens with a trigger question linked to the session's knowledge-test content, and student responses are recorded on a shared board without facilitator evaluation.	A question is posed in the first 10 minutes; responses appear on a visible shared surface (physical or digital); the facilitator does not indicate correct/incorrect responses during this phase.	NID

(Table D1 continued)

No.	Indicator	Observable evidence expected	Component
3	The direct-presentation phase does not exceed 15 minutes.	Observer timestamps the beginning and end of instructor presentation; duration does not exceed 15 minutes within the content-node phase.	NID
4	Each team produces a knowledge node from a differentiated case or dataset and presents it to at least one other team.	Observer can see distinct cases or datasets distributed to different teams; at least one inter-team sharing or rotation occurs before the end of the content-node phase.	NID
5	The scheduled ethical/collaborative component (debate, peer teaching, or co-evaluation, per the weekly rotation) is conducted for at least 12 of the 15 allocated minutes.	The correct rotational component is delivered (verified against the session schedule); it occupies at least 12 minutes of the 60–75 minute phase.	NID

(Table D1 continued)

No.	Indicator	Observable evidence expected	Component
6	The designated AI tool for the session (ChatGPT, Google Colab, or Kaggle) is used as the primary activity during the technological mediation phase.	The tool specified in the session schedule is in use by all teams within 5 minutes of the start of the technological mediation phase; use is substantive (not merely opening the interface).	TM
7	The facilitator adopts a Socratic role during the technological mediation phase: poses questions rather than providing direct answers.	Observer records at least three facilitator–student interactions during this phase; in each, the facilitator responds to a question with a further question or redirection rather than a direct answer.	TM
8	AI tools are positioned as objects of study (not only as productivity aids): at least one activity requires students to analyse, critique, or evaluate the AI output rather than merely use it.	A task prompt or facilitator question is observed that explicitly asks students to assess the accuracy, assumptions, limitations, or ethical implications of an AI-generated output.	TM

(Table D1 continued)

No.	Indicator	Observable evidence expected	Component
9	A GitHub commit of collaborative outputs is completed by each team before the end of the session.	Observer can verify that each team has made at least one commit to their shared repository during the session (facilitator may confirm via screen check at the end of the TM phase).	TM
10	The institutional LMS is used to distribute session materials, host discussion, or collect submissions during the session.	At least one session material is accessed by students via the LMS, or at least one team submits an output through the LMS during the session.	TM
11	The network-connection synthesis is conducted: each team articulates at least one new connection not present at the session's start.	Each team produces a visible or audible statement of a new connection; the facilitator maps connections on a shared surface during the synthesis phase (minutes 75–85).	CA

(Table D1 continued)

No.	Indicator	Observable evidence expected	Component
12	Each student completes a portfolio entry before leaving the session.	The facilitator prompts portfolio entry in the final 5 minutes (minutes 85–90); the observer confirms that students are writing or typing an entry. Spot-checking LMS submissions within 24 hours is reported in the session log.	CA
13	Portfolio entries meet the minimum requirements: at least 80 words and at least one external reference per entry.	Confirmed by facilitator's next-session log (spot-check of LMS submissions). Observer records whether the facilitator communicated the minimum requirements to students at the start of the portfolio phase.	CA

(Table D1 continued)

No.	Indicator	Observable evidence expected	Component
14	When the co-evaluation component is scheduled (Weeks 8–10): the peer-assessment rubric is applied and written feedback is exchanged within the session.	Applicable only in Weeks 8–10; the rubric is visibly in use; teams exchange written (typed or handwritten) feedback before the end of the session. (Scored as Present/Absent; if not applicable to the observed week, scored as Not Applicable and excluded from the denominator.)	CA
15	The network-connection diagram maintained across the 10 weeks is updated during or immediately after the synthesis phase.	A cumulative network diagram is visible in the classroom or on the shared digital board; connections from the current session are added during or immediately after the synthesis phase.	CA

D3. Scoring Rubric

Each of the 15 indicators is rated independently by the observer using a three-point scale. The session fidelity score is the sum of all applicable indicator scores divided by the maximum possible score for that session (15 or 14 if indicator 14 is not applicable).

Table D2: Observer scoring rubric for each indicator.

Rating	Score	Decision rule
Present	1.0	The indicator is fully observed as described. The observable evidence criterion is met without qualification.
Partially present	0.5	The indicator is observed but incompletely. Typical reasons include: the activity occurred but was substantially shorter than specified; fewer than all teams participated; the facilitator partially adopted the expected role but also provided direct answers; or the artefact was produced but did not meet the minimum specification.
Absent	0.0	The indicator is not observed. The expected activity, artefact, or facilitator behavior did not occur.
Not applicable	—	Indicator 14 only, in sessions from Weeks 1–7, when the co-evaluation component was not scheduled. Excluded from the denominator.

Fidelity score calculation.

$$\text{Fidelity score} = \frac{\sum \text{indicator scores}}{\text{number of applicable indicators}} \times 100\%$$

For sessions in Weeks 1–7 (indicator 14 not applicable): denominator = 14, maximum score = 14.

For sessions in Weeks 8–10 (indicator 14 applicable): denominator = 15, maximum score = 15.

D4. Observed Sessions and Fidelity Scores

Four experimental sessions were selected for observation using stratified random sampling. Strata were defined by (a) facilitator (A, B, C, D) and (b) phase of intervention (early: Weeks 1–5; late: Weeks 6–10). One session per facilitator was drawn, with the constraint that at least two sessions fell in each phase.

Table D3: Independent observer session records. Facilitator labels (A–D) are arbitrary within the observation sample.

Obs.	Session	Week	Facil.	Phase 4 component	Applicable indic.	Fidelity
O-1	3	2	A	Structured ethical debate	14 (ind. 14 N/A)	86%
O-2	8	4	B	Structured ethical debate	14 (ind. 14 N/A)	89%
O-3	16	8	C	Portfolio co-evaluation	15	80%
O-4	17	9	D	Portfolio co-evaluation	15	93%
<i>Mean (range)</i>						87% (80–93%)

D4.1. Indicator-Level Scores per Observed Session

Table D4 reports the score assigned to each of the 15 indicators in each of the four observed sessions. “N/A” indicates the indicator was not applicable.

Table D4: Indicator-level fidelity scores across four observed sessions. P = Present (1.0); H = Partially present (0.5); A = Absent (0.0); N/A = Not applicable.

Ind.	Indicator (abbreviated)	O-1	O-2	O-3	O-4
1	Multidisciplinary team composition (4–6 members)	P	P	P	P
2	Trigger question + shared recording, no evaluation	P	P	P	P
3	Direct presentation ≤ 15 min	P	P	H	H
4	Knowledge node produced and shared inter-team	P	H	H	P
5	Phase 4 component ≥ 12 of 15 min	H	P	P	P
6	Designated AI tool in active use within 5 min	P	P	P	P
7	Socratic facilitator role (≥ 3 redirections)	H	P	H	H
8	AI positioned as object of study (critique task)	P	P	P	P
9	GitHub commit completed by each team	A	P	P	P
10	LMS used to distribute/collect during session	P	P	P	P
11	Connection synthesis on shared surface	P	P	P	P
12	Portfolio entry begun before session end	P	H	H	P
13	Minimum requirements communicated (80 words, 1 ref.)	P	H	H	P
14	Peer rubric applied & written feedback exchanged	N/A	N/A	H	P
15	Network diagram updated during/after synthesis	P	P	P	P
<i>Score</i>		<i>12.0/14</i>	<i>12.5/14</i>	<i>12.0/15</i>	<i>14.0/15</i>
<i>Fidelity</i>		<i>86%</i>	<i>89%</i>	<i>80%</i>	<i>93%</i>

Note. The lowest score (O-3, 80%) corresponds to the first session of the portfolio co-evaluation block (Week 8); its partial ratings concentrate in the co-evaluation-related indicators (12–14), consistent with teams’ initial unfamiliarity with the peer-assessment rubric. All partial ratings fell within the ± 3 -minute tolerance of Section D5 and therefore did not constitute formal deviations; the single Absent rating (O-1, indicator 9) corresponds to the Session-3 GitHub deviation recorded in Table D4.

D5. Registered Deviations

Table D5 documents all deviations from the standard session protocol identified through the independent observer records, facilitator session logs, and weekly supervision minutes across the full 20-session sequence. A *deviation* is defined as any instance in which a protocol element was absent, truncated by more than 5 minutes relative to its specified duration, substituted, or reordered. Minor variations within the expected time tolerance (± 3 minutes) are not recorded as deviations.

Table D5: Full deviation log (all 20 experimental sessions).
Source: facilitator session logs and supervision minutes.

Sess.	Wk	Indicator	Deviation	Cause	Facilitator response
3	2	Ind. 5 (Phase 4 duration)	Ethical debate (Case 1) ran 10 min rather than the specified 15 min; teams reached consensus faster than anticipated and the facilitator closed the activity.	Rapid consensus on a relatively straightforward case; group had strong prior agreement.	Facilitator noted in log; added a follow-up question in the next session's trigger (Session 4) to revisit the case.
3	2	Ind. 7 (Socratic role)	In one team interaction, the facilitator provided a direct answer to a technical question about tokenization rather than redirecting.	Student confusion about a concept not yet covered in the content node; facilitator judged that redirection would impede progress.	Discussed at Week 2 supervision; facilitator agreed to use referral to readings or ChatGPT self-exploration for such queries in future.
3	2	Ind. 9 (GitHub commit)	One team did not complete the GitHub commit within the session; submitted within 90 minutes via LMS message to facilitator.	New users unfamiliar with commit workflow; repository configuration issue during setup.	Facilitator sent a brief written guide to the team; issue did not recur for this team in subsequent sessions.

(Table D4 continued)

Sess.	Wk	Indicator	Deviation	Cause	Facilitator response
9	5	Ind. 3 (Presentation $\leq$ 15 min)	Content-node presentation on bias mechanisms extended to approximately 20 min rather than 15 min.	Topic complexity (representation bias, measurement bias, aggregation bias); facilitator chose to provide additional examples to ensure baseline comprehension.	Recalibrated at Week 5 supervision; facilitator agreed to limit future presentations to 15 min and move examples to reading materials.
12	6	Ind. 12 (Portfolio entry begun)	Portfolio entry phase was shortened to approximately 3 minutes (from the specified 5 minutes) because the technological mediation phase ran over by 2 minutes.	Teams were engaged in a data-cleaning task that required more time than allocated; facilitator did not interrupt.	Facilitator assigned portfolio entry as brief follow-up task to be completed within 2 hours via LMS. All entries received within 3 hours of session end (confirmed in log).
13–14	7	Ind. 5 (Phase 4 duration)	The peer-teaching rotation (Week 7, Sessions 13–14, scheduled for 15 minutes) was shortened to approximately 8 minutes due to an unscheduled institutional announcement broadcast through classroom speakers that disrupted the session for approximately 10 minutes.	Unscheduled institutional announcement broadcast through classroom speakers during the activity.	Facilitator documented the disruption; this is the deviation referenced in the main manuscript. No compensatory action was taken; considered a minor scheduling interruption.

(Table D4 continued)

Sess.	Wk	Indicator	Deviation	Cause	Facilitator response
19	10	Ind. 11 (Connection synthesis)	In one section, the connection synthesis was conducted verbally only; the cumulative network diagram was not updated during the session because the projector was unavailable.	Technical failure of the classroom projector at the start of the session.	Facilitator conducted synthesis orally; updated the digital network diagram on the LMS within 24 hours and shared the updated version with the section.

Summary. Seven deviations were recorded across 20 sessions and four facilitators: three in Component 1 (NID; indicators 3, 5, and 5), two in Component 2 (TM; indicators 7 and 9), and two in Component 3 (CA; indicators 11 and 12). None altered the fundamental structure of the intervention. Five of the seven deviations were caused by context-specific factors (unexpected consensus, technical issues, institutional interruption); two reflected facilitator judgement under time pressure. All were identified through facilitator logs or observer records within 24 hours and addressed at the nearest supervision meeting.

D6. Fidelity by Facilitator and Component

Because each of the four facilitators was observed exactly once (one session per facilitator), Table D6 presents both individual facilitator scores and component-level scores from the four observations.

Table D6: Fidelity scores by facilitator (from one observed session each) and by component (mean across applicable observations).

	Facil. A	Facil. B	Facil. C	Facil. D
Observation	O-1 (Sess. 3)	O-2 (Sess. 8)	O-3 (Sess. 16)	O-4 (Sess. 17)
Overall fidelity	86%	89%	80%	93%
NID (ind. 1–5)	90%	90%	80%	90%
TM (ind. 6–10)	70%	100%	90%	90%
CA (ind. 11–15)	100%	75%	70%	100%

Note. Component scores are computed from the applicable indicators within that component in the observed session. CA scores in O-1 and O-2 have a denominator of 4 (indicator 14 N/A); in O-3 and O-4 the denominator is 5. Given one observation per facilitator, component-level scores are descriptive only and should not be interpreted as stable estimates of component-level delivery quality.

D7. Control Session Verification

Two control sessions (Sessions 6 and 14, drawn from different weeks and time slots) were observed by the same independent observer to confirm treatment differentiation. For each session, the observer recorded the presence or absence of eight connectivist elements that, if present, would constitute treatment contamination (Table D7).

Table D7: Treatment-differentiation verification: absence of connectivist elements in two observed control sessions. All eight elements were absent in both sessions (confirmed = Yes).

Element checked for absence	Ctrl. Sess. 6	Ctrl. Sess. 14
Structured multidisciplinary team formation	✓	✓
Peer-teaching rotation	✓	✓
Structured Academic Controversy debate	✓	✓
Mandatory GitHub repository use	✓	✓
Digital portfolio entry task	✓	✓
Network-diagram construction	✓	✓
Peer-assessment rubric application	✓	✓
Kaggle dataset exploration (structured task)	✓	✓

Note. The control instructor used ChatGPT demonstratively during a lecture explanation in Session 6 (not listed above as a checked element, since both conditions had tool access). No structured task requiring students to use ChatGPT was assigned, consistent with the incidental-use specification.